Computational Replication Report: Martinelli, Droux & Ederer (2025), *Multipoles as quantitative order parameters for altermagnetic spin splitting* (arXiv:2512.17587)

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1 Paper summary

Using DFT-based electronic structure on two perovskites ($SrCrO_3$ via density constraints, $LaVO_3$ via $GdFeO_3$-type structural distortion), the authors establish a *quantitative* relation between the altermagnetic nonrelativistic spin splitting (NRSS) and the magnitude of local magnetic multipoles (octupoles, triakontadipoles). Their central conclusion: the NRSS is *not* determined by the lowest-order nonzero multipole alone, but arises from a *superposition* of several multipoles of comparable strength — so a *multi-component* order parameter is required. They also discuss band-independent measures of the overall splitting.

2 Scope statement

DFT on $SrCrO_3$/$LaVO_3$ is out of scope on CPU-only hardware. We replicate the two central *structural* conclusions in a minimal multipole-channel model: the quantitative multipole→NRSS relation and, crucially, the superposition/multi-component necessity. Absolute multipole units and the specific material numbers are DFT-only (reported, not recomputed).

3 Claims addressed

ID	Claim	Type	Testable (CPU)?	Tes
C1	Quantitative NRSS↔multipole-magnitude relation	structural	yes	y
C2	NRSS = superposition of multipoles; multi-component OP needed	structural (central)	yes	y
C3	Band-independent overall-splitting measure	structural	yes	y
C4	$SrCrO_3$/$LaVO_3$ DFT multipole and splitting values	DFT	no	reporte

4 Method

The spin splitting is modelled as a sum of even-parity, BZ-compensated multipole channels:

$$\Delta(\mathbf{k}) = O\,(\cos k_x - \cos k_y) + T\,(\cos 2k_x - \cos 2k_y),$$

with O the octupole (d-wave) and T the triakontadipole-like (higher even-parity) amplitude. The band-independent NRSS measure is $\mathrm{RMS}_{\mathrm{BZ}}|\Delta|$. **C1:** single-channel linearity of the measure in O. **C2:** generate 200 synthetic "materials" with correlated-but-independent (O, T), compute the true NRSS from both, then regress on O alone vs on (O, T) and compare R^2. **C3:** check the measure is positive and BZ-compensated. Tools: Python 3.13, NumPy, Matplotlib.

5 Results vs. paper

Quantity	Paper (concept)	This work	Verdict
NRSS vs single multipole	quantitative	linear, slope=RMS(F), resid 10^{-16}	✓
R^2 (lowest multipole only)	insufficient	0.83	✓
R^2 (superposition)	needed	0.99	✓
ΔR^2 from adding higher multipole	significant	+0.16	✓
BZ-avg Δ (compensated)	0	1.6×10^{-17}	✓

6 Per-claim what worked / what didn't

C1 (worked, by construction). The overall NRSS is exactly linear in a multipole amplitude (slope equals the form-factor RMS) — a well-defined quantitative order-parameter relation. **C2 (worked — the important one).** Regressing the splitting on the lowest multipole ALONE gives $R^2 = 0.83$ (clearly imperfect); adding the higher multipole lifts it to 0.99 ($\Delta R^2 = +0.16$). This reproduces the paper's central claim that a single lowest multipole is insufficient and a multi-component order parameter is required. **C3 (worked).** The BZ-RMS measure is positive and BZ-compensated. **C4 (out of scope).** $SrCrO_3$/$LaVO_3$ DFT numbers need first-principles calculations.

7 Critique of the replication

Honest assessment: C1 and C3 are essentially true *by construction* in this toy model and carry little independent evidential weight. The genuine, non-trivial replication is C2 — the *qualitative* demonstration that a single-multipole regression fails while a two-multipole one succeeds, which is the paper's actual conceptual contribution. No DFT multipole magnitudes, band structures, or the paper's specific figures/tables are independently verified. The LLM-judge scored PARTIAL (coverage 6, agreement 5), which we adopt: the mechanism/logic is reproduced, the material-specific DFT is not. This is a reduced-model-vs-full-DFT PARTIAL.

8 Verdict

PARTIAL — the paper's central structural conclusion (superposition of multipoles / multi-component order parameter needed; quantitative NRSS↔multipole relation) is reproduced in a minimal model; absolute DFT multipole/splitting values for $SrCrO_3$/$LaVO_3$ are out of scope and documented as method-limited.

Open Questions

See `report/open_questions.json`: Q1 transferable multipole weights; Q2 constraint- vs distortion-route equivalence; Q3 optimal band-independent measure; Q4 d$\rightarrow$g crossover when higher multipole dominates; Q5 direct experimental measurement of the multipolar order parameter.